Midterm - INFO7375 Midterm Architectural Report — Option C (Redis)

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We implemented **Option C (ElastiCache Redis INCR Counter)** to protect our order backend, which is strictly capped at 40 requests per second. Redis was chosen because it is explicitly optimized for ultra-fast, atomic operations in highly concurrent environments. Our virtual waiting room acts as an admission gatekeeper positioned securely in front of the POST /orders endpoint.

To pace admissions smoothly without accumulating unused capacity during lulls, the system uses mathematical **Target-Time Pacing**. The handler calculates the exact future timestamp a user is allowed to enter (Target Time = Virtual Start + (Position / Rate)). Visitor state is stored entirely in-memory using a Redis Hash (HSET) to avoid overwhelming DynamoDB with writes during the initial stampede.

When users poll their status, the system checks if the wall-clock time has passed their target time. If it has, they are admitted, and an admission token is written to the DynamoDB Admissions table for the legacy OrderProducer gate to verify and consume. Because Redis executes commands sequentially in a single thread, simultaneous requests are inherently safe from race conditions.

With our Virtual Waiting Room deployed, the system handled the exact same stampede flawlessly, insulating the backend and ensuring 100% fair admission for genuine users.

To satisfy the strict $2.00 AWS exam budget constraint, we purposefully provisioned a single cache.t3.micro ElastiCache node within an isolated VPC instead of a highly available Serverless Cache. While this keeps costs well within limits, it introduces a single point of failure and a hard network bottleneck at the admission layer.

Metric	**Unprotected Baseline**	**Protected Stampede (Option C)**
Total Fans Admitted	1,061 (459 failed)	1,436 (100% success)
Bots Rejected (403)	0 (Bots succeeded)	84 out of 84 (100% blocked)

Metric	Unprotected Baseline	Protected Stampede (Option C)
HTTP 429 Errors	459	0
Wait Times	0s (Instant failure/success)	Med: 64.7s | 95th: 122.7s | Max: 130s